RUDRA SHARMA

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OBJECTIVE

Computer Science (AIML) graduate specializing in building scalable AI applications, high-performance data engineering pipelines, and cloud-based intelligent systems. Proficient in architecting robust backends and optimized processing engines using Python, FastAPI, Apache Spark, and PostgreSQL to deliver production-grade machine learning solutions.

EXPERIENCE

Data Engineering Intern – BytePX March 2026 – Present

- Architected DataPurge Studio SaaS, an asynchronous multi-tenant data engine using FastAPI and PostgreSQL that processes complex multi-sheet workbooks in < 2 minutes with 99.9% accuracy, achieving a 15x˘20x speedup.
- Optimized memory and algorithmic performance by streaming data via io.BytesIO buffers onto Starlette thread pools, implementing dynamic type downcasting, and designing an $O(N)$ linear lookahead sniffer alongside rapidfuzz (C++ layer) for 10x faster fuzzy matching.
- Engineered Databricks & Apache Spark pipelines to automate internal workflows, including a scalable resume converter that eliminated manual parsing for contract tracking, slashing multi-format document processing time by 90%.
- Executed end-to-end client ML delivery by cleaning client datasets, training a Random Forest pipeline integrated with SHAP explainable AI to reduce marketing overhead by 50%, and leveraging IBM watsonx to deploy proof-of-concepts.
- Mentored 2 junior engineering interns on core Python, Apache Spark data frames, and Databricks cluster environments, driving clean-code practices and accelerating feature delivery across 3 internal and client evaluation projects.

AI & ML Intern – UltraTech Cement Ltd. [Birla White] June 2025 – July 2025

- Engineered 'Birbal 2.0', an enterprise AI-powered multilingual chatbot using FastAPI and an LLM-powered Retrieval-Augmented Generation (RAG) pipeline to automate querying across domain-specific technical documentation.
- Slashed employee query response latency by 95% by replacing traditional, blockable SQL/API retrieval sequences with fast, asynchronous NLP-driven data pipelines and intentional text classification algorithms.
- Integrated bilingual (Hindi-English) linguistic support and contextual state tracking, maximizing operational query workflow efficiency by 60% across regional technical teams.

Generative AI Intern – SmartBridge June 2025 – July 2025

- Built and deployed Generative AI architectures using FastAPI and Google Gemini APIs, experimenting with Variational Autoencoders (VAEs), GANs, BERT, and LSTMs.
- Engineered GenAI solutions (multilingual chatbots, predictive models) while completing 60+ hours of advanced training and earning 25+ Google Cloud badges.

SKILLS

Core & Problem Solving: Data Structures & Algorithms (DSA), Python, Java, SQL, JavaScript
ML/AI: Scikit-learn, TensorFlow, PyTorch, XGBoost, NLP, LLMs, RAG, LangChain
Data Engineering: Apache Spark, Databricks, Pandas, ETL Pipelines
Backend & Cloud: FastAPI, Flask, REST APIs, PostgreSQL, Supabase, Git, IBM watsonx
AI Prototyping & Visualization: React, HTML, CSS, Chart.js
Certifications: Microsoft Certified: Azure AI Engineer Associate (Credential: 957CF25F904A0B46); 25+ Google Cloud Skill Badges.
Platform Metrics: 120+ Data Structures & Algorithms problems solved on LeetCode; Java Gold Badge holder on HackerRank.

RESEARCH & PROJECTS

VoxContextEngine – Production Hybrid RAG Platform VOX

- Engineered a document-grounded Retrieval-Augmented Generation (RAG) engine using FastAPI and Docker to enforce context-locked, hallucination-resistant LLM responses.
- Implemented a hybrid search architecture combining dense vector search via Qdrant (all-MiniLM-L6-v2 embeddings with HNSW indexing) and sparse keyword algorithms via Rank-BM25..
- Built an automated Python evaluation suite to benchmark data ingestion and precision metrics, achieving a 100% safety run completeness score during hallucination validation tests.

Estimated Delivery Date Prediction – E-Commerce ML Project EDD-Project

- Trained and optimized an XGBoost predictive model across a dataset of 500K+ historical e-commerce shipment records to enable reliable delivery time predictions.
- Executed advanced feature engineering and data scaling pipelines to capture supply-chain constraints and minimize ETA prediction variance.

Pneumonia Detection & Genomic Analysis – Healthcare ML Medical-ML

- Co-authored an institutional research paper outlining a hybrid deep learning architecture for non-invasive clinical diagnostics.
- Achieved 94% accuracy in automated pneumonia identification by training Convolutional Neural Networks (CNNs) on chest X-ray image datasets.
- Applied logistic regression modeling to complex genomic sequence structures to predict multi-factor polygenic disorder risks.

EDUCATION

University of Petroleum and Energy Studies (UPES) Aug 2022 – May 2026
B.Tech in Computer Science [AIML] *CGPA: 8/10*

St. Anselm's School, Mansarovar, Jaipur May 2019 – Feb 2022
12th CBSE BOARD *93% [PCM]*